THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra (Part 1, 50 marks in total)

May 18, 2022 TIME: 6:30 p.m.-8:30 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer Part I on Moodle directly.

1. (10 Points in total, each question carries 2 marks) Determine whether the following statements are true or false.

   (a) $u_1 = (-1, 2, 4)$ and $u_2 = (5, -10, -20)$ are linearly dependent sets of vectors in R^3.

   (b) $u_1 = (3 - 1)$ $u_2 = (4, 5)$, $u_3 = (-4, 7)$ are linearly dependent sets of vectors in R^2.

   (c) $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ is diagonalizable.

   (d) There are exactly 4 critical points of $f(x, y) = x^3y^4 - x^2y^5$.

   (e) $(0, 0)$ is an saddle point of $f(x, y) = x^2 - y^2 + 2xy$.

2. (40 Points in total, each question carries 4 marks) Short Answers (Use decimal places to represent your answer and round to 2 decimal places if the answer contains more than 2 decimal places.)

   (1) Find all real values of λ do the vectors $v_1 = (\lambda, -\frac{1}{2}, -\frac{1}{2})$, $v_2 = (-\frac{1}{2}, \lambda, -\frac{1}{2})$, $v_3 = (-\frac{1}{2}, -\frac{1}{2}, \lambda)$ form a linearly dependent set in R^3 and add them up.

   (2) Given that $a^2 + b^2 = 1$, calculate the maximum of $(a + 1)(b + 2)$.

   (3) Given that $a^2 + b^2 = 1$, calculate the minimum of $(a + 1)(b + 2)$.

   (4) Using ONE iteration of Newton-Raphson's method with initial guess $x_0 = 1$ to solve the equation $f(x) = \cos(x) - x^3 = 0$.

(5) Find the volume V under the plane $z = 8x + 6y$ over the rectangle R=[0,1]×[0,2].

(6) Determine the area that is inside both $r = 3 + 2\sin\theta$ and $r = 2$.

(7) Determine the area that lies inside $r = 3 + 2\sin\theta$ and outside $r = 2$.

(8) Find the smallest distance from the origin to the intersection of the two planes $y + 2z = 12$ and $x + y = 6$.

(9) Let (x, y) be the center of mass for the triangle with vertices $(0, 0)$, $(-4, 2)$, $(0, 6)$. Calculate $x + y$.

(10) Calculate the minimum of $f(x, y) = \sin x + \sin y - \sin(x + y)$ on $D = \{(x, y)|0 \leq x, y \leq 2\pi\}$.

END OF PART 1

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MATH2014: Multivariable Calculus and Linear Algebra (Part 2, 50 marks in total)

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Answer ALL 5 questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Please arrange and scan your answer sheets in order, from question 1 to question 5.

1. (10 marks) Find the center of mass for the region inside the circle $r = 1$ and outside the circle $r = 2\cos\theta$ (round to 2 decimal places).

2. (10 marks) Use Lagrange multipliers method to find the distance from the line $y = 2x + 3$ to the parabola $y^2 = 2x$. (Round to 2 decimal places; You will not get any marks if you use other methods.)

3. (10 marks) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$.

   (a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

   (b) Is A diagonalizable or not? Why?

4. (10 marks) Compute the determinant for the following matrix. Indicate clearly the method being used.

$$A_k = \begin{bmatrix} 0 & x_1 & x_2 & \dots & x_k \\ x_1 & 1 & 0 & \dots & 0 \\ x_2 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_k & 0 & 0 & \dots & 1 \end{bmatrix}$$

5. (10 marks) Let $v_1, v_2, \cdots, v_k$ be a set of at least two vectors in a vector space V. Show that if one of the vectors in the set is a linear combination of other vectors in the set, then that vector can be deleted from the given set of vectors and the linear span of the resulting set of vectors will be the same as the linear span of $v_1, v_2, \cdots, v_k$.

******* **END OF PART 2** *******